

Introduction to Stata

The Interface, Common Commands, and Exploring Variables

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What is Stata?

Stata is a statistical software package used for:

- ① Data management
- ② Descriptive statistics
- ③ Data visualization
- ④ Statistical analysis/Econometrics/Survey Analysis

The Stata Interface

The five main parts of the Stata interface are:

The screenshot shows the Stata 17.0 interface with the following components highlighted:

- History window:** Located on the left, it shows a list of commands entered, including `sysuse auto`, `codebook mpg`, and `regress price weigh mpg`.
- Results window:** The central window displaying the output of the `regress price weigh mpg` command. It includes a tabulation of unique values for `mpg` and a detailed regression summary table.
- Variables window:** Located on the right, it lists all variables in the dataset with their labels, such as `make` (Make and model), `price` (Price), `mpg` (Mileage (mpg)), etc.
- Properties window:** Also on the right, it shows the properties of the selected variable (`mpg`), including its name, label, type, format, and value labels.
- Command window:** Located at the bottom, it is the area where new Stata commands are entered.

Results window content:

Range: [1,5] Units: 1
Unique values: 5 Missing: 5/74

Tabulation: Freq. Value

Value	Freq.
2	1
6	2
30	3
18	4
11	5
5	.

regress price weigh mpg

Source	SS	df	MS	Number of obs =	74
Model	186325380	2	93162690	F(2, 71) =	14.74
Residual	460744115	71	6489353.7	Prob > F =	0.0000
Total	635069396	73	8699525.97	R-squared =	0.2914
				Adj R-squared =	0.2735
				Root MSE =	2514

	Coefficient	Std. err.	t	P> t	[95% conf. interval]
price	1.746559	.6413538	2.72	0.008	-.467736 3.925382
weight	-49.51222	86.15604	-0.57	0.567	-228.3605 122.278
_cons	1940.009	3597.05	0.54	0.590	-5226.245 9158.382

The Stata Interface

The main parts of the Stata interface are:

- 1 **Command window:**
- 2 **Results window:**
- 3 **Variables window:**
- 4 **Properties window:**
- 5 **History window:**

Can you briefly describe the functions of each?

The Stata Interface

The main parts of the Stata interface are:

- ➊ **Command window:** Interface for command prompts
- ➋ **Results window:** Shows results of your commands
- ➌ **Variables window:** Database of variables in dataset, including names and labels
- ➍ **Properties window:** Listing of variable and dataset properties
- ➎ **History window:** Archive of all commands entered

A typical Stata command has the following structure

```
command object, options
```

Uploading a Dataset

To upload a sample dataset use the following command:

```
sysuse sp500, clear
```


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Describe what you see in the 5 main windows of Stata.

Uploading a Dataset

Now run the following command:

```
browse
```

Describe what you see?

Exploring a Variable: A Systematic Approach

A systematic approach to understanding variables in your dataset is to aim to answer the following questions:

- 1 What does the variable measure and how is it stored?
- 2 What values does it take?
- 3 Is it categorical or numerical?
- 4 How many observations are missing?
- 5 What is its distribution?
- 6 Are there unusual values, possible data-entry errors?

Question 1: What does the variable measure and how is it stored?

To first understand the dataset run the command:

```
describe
```

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To first understand the dataset run the command:

```
describe
```

To focus on a particular variable run the command:

```
describe close
```

Question 2: What values does the variables take?

We can ascertain values of the variable by running the following commands:

```
list close in 1/25
```

```
summarize close
```

Describe what you see...

Question 3 & 4: Is it categorical or numerical and how many observations are missing?

```
codebook close
```

Question 5: What is the variables distribution?

histogram close, frequency normal

- ① What is the Central value?
- ② How widely is the variable spread?
- ③ What is the shape of the distribution (symmetric, right-skewed, left-skewed, bimodal)?
- ④ Where are the peaks?
- ⑤ Are there unusual values?

Question 5: What is the variables distribution?

`histogram close, frequency normal`

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`kdensity close, normal`

Question 5: Are there unusual values, possible data-entry errors?

```
graph box close
```

Question 5: Are there unusual values, possible data-entry errors?

```
graph box close
```

```
graph box close, marker(1, mlabel(date) mlabsize(small))
```

Question 5: Are there unusual values, possible data-entry errors?

```
graph box close
```

```
graph box close, marker(1, mlabel(date) mlabsize(small))
```

```
twoway line close date, title(Closing Price Per Day)
```

Exploring Your Data Further with Comparison Operators

Important comparison operators include:

Operator & Meaning

`==` & Equal to

`!=` & Not equal to

`>` & Greater than

`<` & Less than

`>=` & Greater than or equal to

`<=` & Less than or equal to

Example:

```
summarize close if date >= mdy(4,15,2001)
summarize close if date < mdy(4,15,2001)
```

Logical Operators

Stata uses:

Operator	Meaning
&	AND
	OR
!	NOT

Example:

```
summarize close if date >= mdy(4,15,2001) & date <= mdy(5,15,2001)
```

Average closing price between 15th April and 15th May

Replication: Introduction to Do files

A **do-file** is a text file containing a sequence of stata commands that can be executed together. To access the do file type the following in a command window:

```
doedit
```

Replication: Introduction to Do files

Exercise 1:

Carefully copy and paste all the codes that we have used so far into the do file (after the commands **cls** and **clear all**) and run them by clicking on Execute/do:

```
cls
clear all
sysuse sp500, clear
describe close
list close in 1/20
summarize close
summarize close, detail
codebook close
histogram close, xlabel(0(5)80) frequency normal
kdensity close, normal
graph box close, marker(1, mlabel(date) mlabsize(small))
twoway line close date, title(Closing Price Per Day)
```


Replication: Why do we use do-files?

- ① **Reproducibility:** You can run the same analysis again
- ② **Organization:** All commands for an analysis are kept in one place
- ③ **Error Correction:** By simply modifying commands
- ④ **Documentation:** Record of how results were produced
- ⑤ **Efficiency:** Enable the running of multiple codes rather than typing them repeatedly

Replication: Creating Beautiful Codes with Comments in Stata

Stata provides three common ways to add comments to your code.

Method	Example	Best Use
//	// This is a comment	Single-line comments
/* */	/* Several lines */	Multi-line comments
*	* This is a comment	Simple headings

Example:

- // Load the data
- /* Explain several lines of code */
- * DATA TRANSFORMATION

Replication: Creating Beautiful Codes With Spacing and Comments

Exercise 2:

```
lecture_1_explore_data X
1 *****
2 *Project: Explore Data on Closing Prices on the S&P500
3 *Author: Lecturer
4 *Date: August 17th 2026
5 *****
6
7
8 *1. Clear the workscreen and workspace
9 cls
10 clear all
11
12 *2. Load the Data
13 sysuse sp500, clear
14
15
16 *****
17 *Learn as much as you can about the expenditure variable
18 *****
19
20 *3. Find out what the expenditure variable measures and how it is stored
21 describe close
22
23
24 *4. What values does it take?
25
26 list close in 1/25
27 summarize close
28 summarize close, detail
29
30 *5. Is it categorical or numerical and how many observations are missing?
31
32 codebook close
33
34 *6. What is its distribution
35
36 histogram close, xlabel(0(5)80) frequency normal
37 kdensity close, normal
38
39 *7. Are there unusual values?
40
41 graph box close, marker(1, mlabel(date) mlabsize(small))
42 twoway line close date, title(Closing Price Per Day)
43
44
45
46 exit
```

Replication: Documenting your Codes and Reports

Log your Code Files to create a full record of what you asked Stata to do and what it produced

```
capture log close
log using FileLocation/lecture1.log, replace
...
...
...
log close
exit
```

Why are log files important?

Log Files:

- ① Provide a record of your analysis
- ② Make your research reproducible
- ③ Allow a review of your work
- ④ Provide an audit trail

Exercise 3

Considering the volume variable in the sp500 dataset answer the following questions

- ① What does the variable measure and how is it stored?
- ② What values does it take?
- ③ Is it categorical or numerical?
- ④ How many observations are missing?
- ⑤ What is its distribution?
- ⑥ Are there unusual values, possible data-entry errors?
- ⑦ In one paragraph or so what are your key findings on the volume variable

Recap

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Lesson today is in line with the first stage of analysis

Which is not to go straight to analysis but rather to get to know your data and variables thoroughly.

Exercise 4

Formation of Groups:

- ① Group 1: Revenue Group
- ② Group 2: Expenditure Group
- ③ Group 2: Overall Balance Group

See Group Exercise 1 Document